

TABLE 2
CAPABILITY CONTAINER FILE CONFIGURATION

Block	Byte	Value (Hex)	Description
Block 0	byte 0	E2	Magic number (NFC Forum type 5 identifier)
Block 0	byte 1	40	Mapping version 1.0
Block 0	byte 2	00	Memory size (MSB)
Block 0	byte 3	01	Memory size (LSB)
Block 1	byte 4	00	Read access (no restriction)
Block 1	byte 5	00	Write access (no restriction)
Block 1	byte 6	04	NDEF area start block
Block 1	byte 7	00	RFU (reserved for future use)

is then ready for NFC configuration and programming.

The assembly tools and materials include a 10cc solder paste syringe, flux, SMD component reels, and an A4

self-healing cutting mat, providing a clean and organised work surface. After programming, the completed NFC business card can instantly share contact information, website URLs, digital business cards (vCards), or social media profiles with an NFC-enabled smartphone by simply tapping the card.

NFC configuration and programming

The NFC IC must now be configured and programmed with the required contact details. The free ST25 NFC Tap app is available from Google Play Store (Android) as well as Apple App Store (iOS). After the app is installed, tap the NFC business card against an NFC-enabled smartphone. The app detects the ST25DV64K chip and displays its memory status. If the IC is new, the NDEF area will not be initialised and must be formatted before data can be written.

What is NDEF?

NDEF (NFC data exchange format) is the standard defined by the NFC forum for storing NFC data. It enables smartphones to automatically read information such as a vCard, website URL, phone number, email address, or plain text without requiring an additional app.

After the NFC card is detected, open the memory or areas content editor and select Area 1, which is the main user memory area. Next, configure the capability container (CC) file. The CC file is a small header located at memory address 0x0000 that identifies the tag as an NFC forum Type 5 device and specifies the available memory size and NDEF support. Without a valid CC file, most smartphones will not recognise the tag as an NDEF-enabled NFC card.

Open manual write (raw memory) mode and write the following eight bytes at address 0x0000:

E2 40 00 01 00 00 04 00

This configuration enables

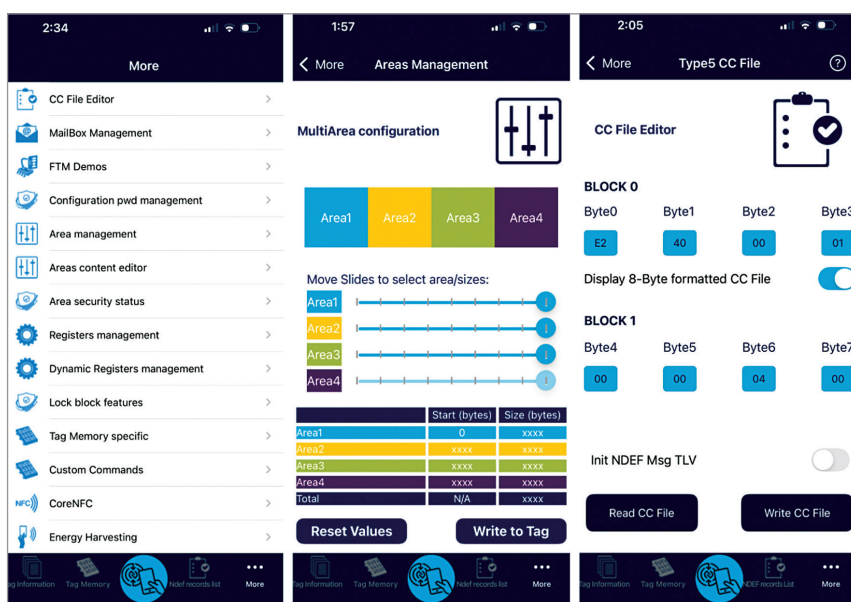


Fig. 7: NFC tap app menu configuring the NDEF memory area and CC file configuration

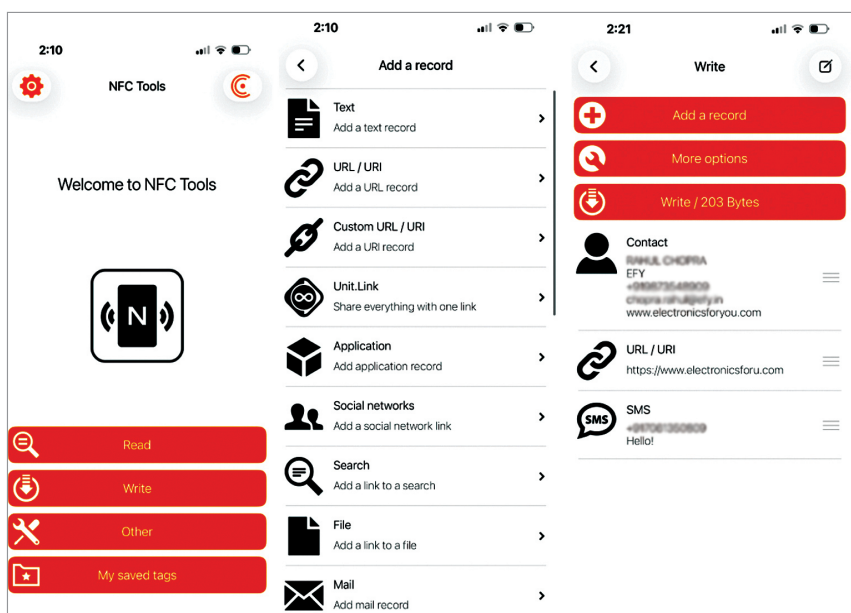


Fig. 8: Writing the VCard data to the NFC IC of the business card